Exercise Set 4 - Part 1

Problem 1

1.1 Give two advantages of panel data.

- Controlling for individual heterogeneity (e.g. unobserved time-invariant variables such as ability or motivation)
- More informative data due to more variability (i.e. variability within and between units)
- Analyze the dynamics of adjustment

1.2 Give two limitations of panel data.

- Design and data collection problems (e.g. frequency of interviewing, interview spacing)
- Distortions of measurement errors (e.g. due to recall, learning effects)
- Selectivity (e.g. attrition)
- Short time-series dimension

1.3 What are the two basic approaches to model unobserved effects in panel data? How do the two approaches differ (assumptions, advantages/disadvantages)?

- Fixed effects and random effects are the two basic approaches.
- Random effects approach assumes that the individual-specific error term component and the explanatory variables are uncorrelated.
- The fixed effects modelling has a general approach and it produces robust estimator; the fixed effects estimator is consistent even if random effects approach is the right one. However, it is inefficient if random effects model is the right one. In addition, fixed effects model precludes time invariant variables.
- The random effects model allows time invariant parameters, has an efficient estimator using GLS, and has lower standard errors than fixed effects model. One of the disadvantages of the random effect model is the assumption that the individual specific error component and the explanatory variables are uncorrelated. If this assumption fails, the random effects estimator becomes inconsistent.

Problem 2

Using the Youth Sample of the National Longitudinal Survey, log wages are explained from years of schooling, years of experience and its square, dummy variables for being a union member, working in the public sector and being married and two racial dummies. The estimation results under three different models are reported below.

2.1 If the assumptions of the random effects model are satisfied, which estimators in Table 1 are consistent? Which estimator is the most efficient one?

	OLS	Fixed Effects	Random Effects
schooling	0.099	-	0.101
	(0.005)	-	(0.009)
experience	0.089	0.116	0.112
	(0.010)	(0.008)	(0.008)
$experience^2$	-0.0028	- 0.0043	-0.0041
	(0.0007)	(0.0006)	(0.0006)
union member	0.180	0.081	0.106
	(0.017)	(0.019)	(0.018)
married	0.108	0.045	0.063
	(0.016)	(0.018)	(0.017)
black	-0.144	-	-0.144
	(0.024)	-	(0.048)
hispanic	0.016	-	0.020
	(0.021)	-	(0.043)
public sector	0.004	0.035	0.030
	(0.037)	(0.039)	(0.036)
constant	-0.034	-	-0.104
	(0.065)	-	(0.111)

Table 1: Dependent variable: log(wage), Standard errors are in parentheses

- All three estimators are consistent.
- The random effects estimator is the most efficient one.

2.2 If the individual-specific error term component is correlated with one or more of the explanatory variables, which estimators in Table 1 are consistent? Which test would you perform to check this hypothesis?

- The fixed effects estimator is the only one that is consistent.
- Hausman test

2.3 Show the within transformation of the panel data model $y_{it} = \alpha_i + \mathbf{x}_{it}'\beta + \varepsilon_{it}$. What assumption is necessary to make the fixed effects estimator consistent?

$$(y_{it} - \bar{y}_i) = (\alpha_i - \alpha_i) + (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i),$$

where $\bar{y}_i = T^{-1}\sum_t y_{it}$. (Note that α_i drops out.)

If we assume $E[(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\varepsilon_{it}] = \mathbf{0}$, then $\hat{\beta}_{FE}$ is consistent (for $N \to \infty$ and fixed T).

2.4 Compare the results of fixed effects and random effects model. Interpret the coefficient of being married.

- The fixed effects approach eliminates any time-invariant variables from the model. The effects of schooling and race are wiped out. The random effects approach column presents the GLS estimator and time-invariant variables are also estimated.
- The differences among estimates are not substantial. Only being a union member and marital status lead to have higher estimates in the random effects model.
- Marital status is a variable that is likely to be correlated with the unobserved heterogeneity in the individual-specific term. The marital dummy is typically capturing other (unobservable) differences between married and unmarried workers. When the model is estimated by fixed effects approach, the effect of being married reduces to 4.5 %. The effect of being married or being a labour union member in the fixed effects approach is identified only through people who change status over the sample period.

Problem 3

Utilizing panel data on individuals from the NLSY, we will analyze the impact of education on log wages. We include dummy-coded variables for region, race, sex and marital status as well as experience (number of years in the workforce) and tenure (number of years at the specific job), both also squared.

3.1 Set data as panel data.

```
d<- read.csv("YOUR_PATH/NLSY.csv", header=TRUE, sep=",")
d <- pdata.frame(d, index=c("pid","wave"))
pdim(d)

> Unbalanced Panel: n = 7914, T = 1-10, N = 44037
```

3.2 Run a pooled OLS regression. What is the return of education? Assess whether it is statistically significant.

```
pols <- lm(lhw ~ edu_c+tenure + tenure2 + workexp + workexp2 + race1 + race2 +
        marital1 + marital3 + marital4 + marital5 + sex1 +region1 + region2 + region3, data=d)

round(summary(pols)$coefficients, 4)
```

- The individual will earn, all other things equal, approx. 7 % more if education increases by one unit (year). More precisely, the gain will be exp(0.0679)-1*100%=7.03%.
- It is statistically significant at all conventional significance levels.

3.3 Include individual-level fixed effects.

```
fe <- plm(lhw ~ edu_c+tenure + tenure2 + workexp + workexp2 +
        race1 + race2 + marital1 + marital3 + marital4 + marital5
        + sex1 +region1 + region2 + region3,
        data = d, model = "within")

round(summary(fe)$coefficients, 4)
```

Including fixed effects controls for individual unobservables that impact both education and log wages. We control for both ability and personality traits.

3.4 Estimate a random effects model.

```
re <- plm(lhw ~ edu_c+tenure + tenure2 + workexp + workexp2 + race1 + race2
          + marital1 + marital3 + marital4 + marital5 + sex1 +region1 + region2 + region3,
          data = d, model = "random")

round(summary(re)$coefficients, 4)
```

3.5 Interpret and compare the estimates of the three estimated models.

```
export_summs(pols, fe, re, model.names=c("POLS", "FE", "RE"))
```

	POLS	FE	RE
(Intercept)	0.94 ***		0.85 ***
	(0.01)		(0.02)
edu_c	0.07 ***	0.09 ***	0.08 ***
	(0.00)	(0.00)	(0.00)
tenure	0.00 ***	0.00 ***	0.00 ***
	(0.00)	(0.00)	(0.00)
tenure2	-0.00 ***	-0.00 ***	-0.00 ***
	(0.00)	(0.00)	(0.00)
workexp	0.06 ***	0.08 ***	0.08 ***
	(0.00)	(0.00)	(0.00)
workexp2	-0.00 ***	-0.00 ***	-0.00 ***
	(0.00)	(0.00)	(0.00)
race1	0.01		0.01
	(0.01)		(0.01)
race2	-0.02 ***		-0.04 ***
	(0.01)		(0.01)
marital1	-0.13 ***	-0.09 ***	-0.11 ***
	(0.01)	(0.01)	(0.01)
marital3	-0.07 *	-0.04	-0.06 *
	(0.03)	(0.03)	(0.03)
marital4	-0.05 **	-0.04 *	-0.04 *
	(0.02)	(0.02)	(0.02)
marital5	-0.27 **	-0.10	-0.14
	(0.10)	(0.11)	(0.10)
sex1	0.14 ***		0.14 ***
	(0.00)		(0.01)
region1	-0.02 *	-0.05 *	-0.03 *
	(0.01)	(0.02)	(0.01)
region2	-0.11 ***	-0.13 ***	-0.12 ***
	(0.01)	(0.02)	(0.01)
region3	-0.12 ***	-0.10 ***	-0.11 ***
	(0.01)	(0.02)	(0.01)
N	44037	44037	44037
R2	0.19	0.18	0.22

*** p < 0.001; ** p < 0.01; * p < 0.05.

3.6 Run a Hausman test for fixed versus random effects model. What is your conclusion?

```
phtest(fe,re)

Hausman Test
data:  lhw ~ edu_c + tenure + tenure2 + workexp + workexp2 + race1 +  ...
chisq = 500.55, df = 12, p-value < 2.2e-16
alternative hypothesis: one model is inconsistent

We have significant result (p-value=0.000).  The models differ, thus we need to use the fixed
effects model.
```